

Shibji Shekhar Rout

AI Engineer | RAG | LangGraph | Multi-agent Orchestration | Azure

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PROFESSIONAL SUMMARY

AI Engineer with **3 years of enterprise delivery** and an **MSc in Advanced Computer Science** (Leeds, Merit), owning **6+ client relationships** across finance and tech, transitioning from backend systems into AI engineering. Built and deployed production **LangGraph** pipelines and **RAG** systems on **Azure**, achieving **98% pipeline success rate**.

Skills Summary

Generative AI & LLMs: LangChain, LangGraph, LangSmith, OpenAI, Retrieval-Augmented Generation (RAG), Prompt Engineering, MCP, Graph Memory

Vector Databases & Retrieval: Qdrant, ChromaDB, Azure AI Search, Embeddings, Hybrid Search, Re-ranking

Languages & Frameworks: Python, FastAPI, Flask, Asyncio, SQL, REST APIs

Machine Learning: Scikit-Learn, XGBoost, CatBoost, PyTorch, Pandas, NumPy, RAGAS

Cloud & DevOps: Azure (App Services, Video Indexer, ACR, Application Insights), Docker, GitHub Actions, CI/CD

Databases & Messaging: MongoDB, Redis, Upstash, RQ

EDUCATION

University of Leeds

Leeds, UK

MSc in Advanced Computer Science | Grade: Merit

Sep 2024 – Nov 2025

- **Relevant Modules:** Machine Learning, Deep Learning, Cloud Computing, Data Mining, Data Science, Algorithms

PROJECT EXPERIENCE

Brand Guardian AI: Multi-modal Video Compliance Auditing Engine | [Link]

LangGraph · Azure Video Indexer · Azure AI Search · OpenAI · FastAPI · Docker · LangSmith

- Architected a **4-node multi-modal** compliance pipeline using **LangGraph**, orchestrating video ingestion, OCR, speech-to-text and **GPT-4o** auditing against real **FTC** and **ASA/CAP Code** disclosure rules
- Built on full **Azure cloud infrastructure** (**Blob Storage**, **Azure Video Indexer**, **AI Search Vector DB**) and **FastAPI** backend, along with clean separation across entry, orchestration and intelligence layers
- Achieved **98% pipeline success rate** across all production runs at **\$0.001–\$0.004 per audit**, with dual observability via **LangSmith** tracing and **Azure Application Insights** logs and metrics
- Deployed on **Azure**; resolved **YouTube server-side blocking** via direct video upload fallback

DocLense | [Link]

FastAPI · LangGraph · OpenAI · Qdrant · MongoDB · Redis · Docker · Azure

- Architected a production **RAG** system with **FastAPI** across two decoupled pipelines — async PDF ingestion via **RQ** and **Upstash Redis** into **Qdrant Cloud**, and a real-time **LangGraph** chat flow — achieving Context Precision **1.00**.
- Engineered a **5-node LangGraph** pipeline with **token-budgeted MongoDB** history, routing queries through **gpt-4o-mini** first, escalating to **gpt-4o** only on evaluator rejection, with Faithfulness **0.93** across 10 mixed real-world documents.
- Containerised server and worker with **Docker**, deploying as two independent **Azure Container Apps** via **Azure Container Registry**, with ingestion and inference scaling separately through **Azure File Share** inter-container storage.

Student Performance Predictor | [Link]

Python · Scikit-Learn · XGBoost · CatBoost · Flask · Docker · Azure · GitHub Actions

- Conducted **multivariate EDA** across 5 socioeconomic features, identifying lunch type and parental education as statistically significant math score predictors
- Benchmarked **9 regression models** via **GridSearchCV**, selecting **Ridge Regression (88.06% R²)** — catching Decision Tree overfitting at **99.97% train vs 75.72% test**
- Engineered **modular ML pipeline** with ingestion, transformation and trainer components and deployed live on **Azure**

PROFESSIONAL EXPERIENCE

Accenture

Gurgaon, India

Application Development Analyst

Jul 2021 – Jul 2024

- Sole **technical point of contact** for **6+ enterprise clients** across finance and tech, independently owning **end-to-end delivery** and stakeholder communication
- Diagnosed production failures via systematic **JDE log analysis**, drove **architectural upgrade from JDE 9.1 to 9.2** — reducing P1 incidents by **75%**
- Engineered **SQL solutions** processing **1M+ financial records** on live production systems, including **INR currency integration** directly impacting client financial operations
- Led and trained a **team of 2 junior analysts** on production debugging, JDE log analysis and client communication — driving independent delivery within **3 months**
- Cleared **TechLeap AI/ML and Python assessments**, securing fast-tracked promotion within **first year** at Accenture